CognitiveOS — Full Feature Set & Status

Current architecture and qualification status · consolidated implementation inventory

Legend: ■ implemented / qualified ■ final hardening / qualification

Cognitive Core

■ Natural-language intent ingestion

■ Intent → Goal transformation

■ World-grounded planning

■ Execution graph generation

■ Plan validation

■ Dependency reasoning

■ Ordered execution

■ Independent action decomposition

■ Constraint-aware planning

■ Priority-aware planning

■ Preference-aware planning

■ Prediction before execution

■ Observe → Compare → Learn loop

■ Belief update

■ Replanning after world change

World Model & Grounding

■ Persistent world representation

■ Current world-state grounding

■ Entity resolution

■ Capability grounding

■ Provider grounding

■ Availability reasoning

■ Inventory reasoning

■ World mutation handling

■ Stale-state detection

■ World-state validation

■ World-change observation

■ Belief/world consistency checking

Execution Runtime

■ CognitiveRuntime

■ Plan / Observe / Act / Learn loop

■ Execution Graph

■ Execution state tracking

■ Execution node dependencies

■ Action dispatch

■ Capability discovery

■ Capability execution

■ Execution result handling

■ Execution provenance

■ Correlation tracking

■ Causation tracking

- ■ Generic recovery contract
- ■ Retry execution
- ■ Same-tick recovery
- ■ Partial-failure handling
- ■ Dependency-failure handling

Domain Isolation

- ■ Dependency injection
- ■ One-way capability registration seam
- ■ Domain capabilities register into CognitiveOS
- ■ No OS imports of grocery implementations
- ■ ActionExecutor does not inspect domain parameters
- ■ Domain parameters remain opaque to OS runtime
- ■ Domain-specific recovery remains in capabilities
- ■ Non-grocery domain compatibility
- ■ Fictional machine domain verified
- ■ Fictional robotics domain verified
- ■ Static OS/domain boundary guard
- ■ Behavioral domain-isolation tests
- ■ 13/13 isolation tests passing
- ■ 84/84 deterministic regression tests passing

Failure & Recovery

- ■ Deterministic fault injection
- ■ Provider failure detection
- ■ Capability failure detection
- ■ Failure propagation
- ■ Failure comparison
- ■ Recoverable failure classification
- ■ Generic retry
- ■ Alternative-provider recovery
- ■ Same-tick recovery
- ■ Partial recovery
- ■ No duplicate successful actions
- ■ Recovery provenance
- ■ Recovery boundedness
- ■ Safe generic fallback

Human-in-the-Loop

- ■ Human approval requirement
- ■ Approval boundary
- ■ WAITING_FOR_HUMAN state
- ■ Persistent pending approval
- ■ Human approval
- ■ Human rejection
- ■ Resume original execution
- ■ Approval survives restart
- ■ No unauthorized action during pause
- ■ No duplicate action after resume

- ■ Live HTTP E2E verification

Persistence & Checkpointing

- ■ Execution checkpoints
- ■ Persistent execution state
- ■ Runtime restart recovery
- ■ Pending execution restoration
- ■ Completed-node preservation
- ■ Pending-node preservation
- ■ Resume from checkpoint
- ■ Duplicate-action prevention
- ■ Checkpoint goal preservation
- ■ Original request preservation
- ■ Live checkpoint/resume verification

Multi-Agent Runtime

- ■ Multiple cognitive actors
- ■ Actor identity
- ■ Actor-specific execution state
- ■ Actor isolation
- ■ Concurrent actors
- ■ Conflicting beliefs/observations
- ■ Shared goals
- ■ Delegation
- ■ AskActor
- ■ Delegated execution
- ■ Delegated result propagation
- ■ Delegation fallback
- ■ Negotiation
- ■ Actor-to-actor coordination
- ■ Correlated delegated transactions
- ■ Causation/provenance

Shared Resources

- ■ Shared resource representation
- ■ Shared budget
- ■ Cross-actor budget accounting
- ■ Budget reservation
- ■ Budget reconciliation
- ■ Concurrent budget enforcement
- ■ Failed-transaction budget recovery
- ■ Shared constraint enforcement
- ■ Live shared-budget verification

Learning

- ■ Prediction
- ■ Execution outcome capture
- ■ Expected vs actual comparison
- ■ Comparator
- ■ Learning evidence generation

- Success learning
- Failure learning
- Provider performance learning
- Persistent learned state
- Cross-execution learning
- Learning inspection
- Learning from actual outcomes
- Historical evidence preservation

Policy & Governance

- Policy enforcement
- Runtime policy
- Security policy
- Learning policy
- Fraud-review gate
- Fraud-risk evaluation
- Velocity/cooldown visibility
- Permission enforcement
- Communication permissions
- Policy-aware execution
- Security controls preserved during qualification
- No policy bypass

Security

- Identity
- Authorization
- Delegation
- Consent
- Revocation
- Private cognition
- Pre-commit security
- Negotiation-before-commit
- TransitionGate
- World commit authorization
- Mutation protection
- Execution audit
- Actor attribution
- Policy audit
- Consent audit
- Negotiation audit
- Transition audit
- Capability audit
- Security audit
- Security monitoring
- Security violation detection
- Cross-actor access monitoring
- Security dashboard

Trust, Membership & Society

- ■ Society model
- ■ Actor membership
- ■ Actor Registry
- ■ Agent Registry
- ■ Capability Registry
- ■ Provider Registry
- ■ Membership scoping
- ■ Membership revocation
- ■ Shared World
- ■ Trust network
- ■ Affiliation Graph
- ■ Governance
- ■ Policy registry
- ■ Society-level capabilities
- ■ Society administration
- ■ Trust relationships
- ■ Actor affiliation
- ■ Provider affiliation
- ■ Trust-aware discovery

Agents & Communication

- ■ Agent model
- ■ Agent identity
- ■ Agent registry
- ■ BrocaAgentRegistry
- ■ Agent discovery
- ■ AgentBus
- ■ Agent routing
- ■ Actor-to-agent communication
- ■ Agent-to-agent communication
- ■ Sender attribution
- ■ Receiver attribution
- ■ Conversation history
- ■ Communication isolation
- ■ Conversation security
- ■ Agent authorization

Capabilities

- ■ Capability model
- ■ CapabilityRegistry
- ■ Capability discovery
- ■ CapabilityBus
- ■ Capability routing
- ■ Capability execution
- ■ Capability/provider separation
- ■ Capability authorization
- ■ Capability scope
- ■ Capability security

- ■ Capability observability
- ■ Capability → provider routing

Providers

- ■ Provider model
- ■ ProviderRegistry
- ■ Provider discovery
- ■ ARD discovery
- ■ Local registry discovery
- ■ OpenClaw provider
- ■ N8n provider
- ■ NANDA fallback
- ■ Provider selection
- ■ Provider → capability mapping
- ■ Provider execution
- ■ Provider health
- ■ Provider observability
- ■ Provider security/policy
- ■ Provider admin UI

Negotiation

- ■ Negotiation model
- ■ NegotiationPlanner
- ■ TransactionCoordinator
- ■ PendingNegotiation
- ■ Contention detection
- ■ Counterparty identification
- ■ Negotiation lifecycle
- ■ Accept/reject
- ■ Negotiation state
- ■ Negotiation isolation
- ■ Negotiation before commit
- ■ Negotiation → TransitionGate
- ■ Negotiation admin UI
- ■ Negotiation trace

Commerce / Reference Grocery Domain

- ■ Product selection
- ■ Provider selection
- ■ Order creation
- ■ Payment confirmation
- ■ Payment
- ■ Order confirmation
- ■ Delivery
- ■ End-to-end transaction
- ■ Real-world inventory
- ■ Real-world availability
- ■ Real execution state
- ■ Wallet

- ■ Actor orders
- ■ Actor preferences
- ■ Order → execution linkage
- ■ Order → security linkage
- ■ Order → negotiation linkage

Edge / Cloud Deployment

- ■ Edge CognitiveOS
- ■ Cloud CognitiveOS services
- ■ Per-actor process
- ■ Per-actor PID/log
- ■ Edge startup
- ■ Edge shutdown
- ■ Edge actor deployment
- ■ Kubernetes deployment
- ■ Actor-specific pods
- ■ Shared Society infrastructure
- ■ Edge → Cloud sync
- ■ Actor-specific state
- ■ Deployment scripts
- ■ Actor deployment observability
- ■ Durable production recovery
- ■ Large-scale load qualification

Observability & Admin Console

- ■ Lemon metrics
- ■ Runtime metrics
- ■ Execution metrics
- ■ Pipeline metrics
- ■ Actor monitoring
- ■ World monitoring
- ■ Security monitoring
- ■ Execution trace
- ■ Debugger
- ■ Plan Analyzer
- ■ Context visualization
- ■ Grounding visualization
- ■ Audit visualization
- ■ Provider monitoring
- ■ Dashboard
- ■ Actors
- ■ Societies
- ■ World Map
- ■ Timeline
- ■ Negotiations
- ■ Knowledge Graph
- ■ Grounding Graph
- ■ Context Stream

- ■ Memories
- ■ Affiliations
- ■ Capabilities
- ■ Providers
- ■ Communication
- ■ Security
- ■ Orders & Wallet
- ■ Settings

Verification & Qualification

- ■ Unit test triage
- ■ Integration — 85/85
- ■ Deterministic solvers — 61/61
- ■ Qualification regression — 10/10
- ■ Actor isolation — 10/10
- ■ State-transition gate — 8/8
- ■ Milk E2E with strict ordering
- ■ Frontend qualification
- ■ Security qualification
- ■ Deployment verification
- ■ Provider tests
- ■ Negotiation tests
- ■ Remaining infra-timing qualification
- ■ Remaining architecture/constitution cleanup

Benchmark Scenarios

- ■ MB-0001 Hello World / Milk — PASS
- ■ MB-0002 Multi-Agent shared budget — PASS
- ■ MB-0003 Provider Discovery — PASS
- ■ MB-0004 Budget Constraint — PASS
- ■ MB-0005 Priority Constraint — PASS
- ■ MB-0006 Reservation — PASS
- ■ MB-0008 Interruption — PASS
- ■ MB-0009 Temporal reasoning — PASS
- ■ MB-0010 Negotiation — PASS
- ■ MB-0011 Belief vs Reality — PASS
- ■ MB-0012 Human approval — PASS
- ■ MB-0013 Long-running plan — PASS
- ■ MB-0014 Partial failure — PASS
- ■ MB-0015 Learning — PASS

Completion by Domain

DOMAIN	COMPLETION
CognitiveOS runtime	**98%**
Cognitive loop	**97%**
World model	**95%**
Memory / beliefs	**97%**
Agents	**95%**

DOMAIN	COMPLETION
Capabilities	**95%**
Providers	**95%**
Society	**95%**
Trust / affiliations	**95%**
Negotiation	**95%**
Governance	**95%**
Security	**100% — qualified**
Commerce	**95%**
Edge/cloud	**90%**
Observability	**95%**
Admin frontend	**95%**
Testing / qualification	**90%**
Production hardening	**80–85%**
OVERALL	**94–95%**

Qualification Status

AREA	STATUS
Core CognitiveOS	**IMPLEMENTED + DETERMINISTICALLY QUALIFIED**
Domain Isolation	**IMPLEMENTED + ARCHITECTURALLY VERIFIED**
Human Approval	**IMPLEMENTED + LIVE E2E VERIFIED**
Recovery	**IMPLEMENTED + VERIFIED**
Multi-Agent Coordination	**IMPLEMENTED + VERIFIED**
Checkpoint / Restart	**IMPLEMENTED + LIVE VERIFIED**
Shared Resources	**IMPLEMENTED + LIVE VERIFIED**
Learning	**IMPLEMENTED + VERIFIED**
Governance	**IMPLEMENTED + VERIFIED**
Security	**IMPLEMENTED + QUALIFIED**
Providers	**IMPLEMENTED + VERIFIED**
Admin Frontend	**IMPLEMENTED + QUALIFIED**
Full Feature Set	FINAL QUALIFICATION REMAINS

Overall completion: approximately **94–95%** feature/architecture complete.

Remaining work: production hardening, architecture-boundary cleanup, large-scale/parallel qualification, durable recovery, and final qualification evidence.

Project phase: Qualify, harden, package, and deploy the CognitiveOS.